Complex Analysis

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BOSTON UNIVERSITY

MA713

Contents

Abstract

MA713: Functions of a Complex Variable I is a graduate level course in single variate complex analysis, taught by Professor David Rohrlich in the spring 2026 semester (my junior year). The main text we followed was [GK25].

Chapter 1

Fundamental concepts

1.1 Holomorphic functions

Lecture 1: Syllabus day

Tue 20 Jan 2026 09:29

the hw is calc 2. radius of convergence is 1 for both

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \qquad \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}.$$

seems relevant for hw maybe?

Recall that $\mathbb{R}^2 \cong \mathbb{C}$ as $\mathbb{R}$-vector spaces via $(1,0) \mapsto 1$ and $(0,1) \mapsto i$. When is an $\mathbb{R}$-linear map $L : \mathbb{R}^2 \to \mathbb{R}^2$ $\mathbb{C}$-linear when viewed as a map $\mathbb{C} \to \mathbb{C}$? Recall $\dim_{\mathbb{R}} \mathbb{C} = 2$ and $\dim_{\mathbb{C}} \mathbb{C} = 1$. Well a $\mathbb{C}$-linear map $L : \mathbb{C} \to \mathbb{C}$ from a 1-dimensional space to itself is fully determined by the image $L(1)$, since $L(z) = zL(1)$. So an $\mathbb{R}$-linear map $L : \mathbb{R}^2 \to \mathbb{R}^2$ is $\mathbb{C}$-linear precisely when $L(0,1) = (0,1)L(1) = (0, L(1))$, where $L(1,0)$ is of the form $(\alpha, 0)$ and we define $L(1) \coloneqq \alpha$.

This is a shitty characterization. We can try to characterize the matrix of L as an $\mathbb{R}$-linear map. We start with a map $L : \mathbb{C} \to \mathbb{C}$. As a map $\mathbb{R}^2 \to \mathbb{R}^2$ the matrix is simply given as follows. Take $L(1) + a + bi$. Then the matrix of L is $L(1,0) = (a,b)$ and $L(0,1) = i(a+bi) = -b + ai = (-b, a)$. Thus the matrix relative to the standard basis is

$$L \sim \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

Thus $\mathbb{R}$-linear maps which induce $\mathbb{C}$-linear maps are precisely matrices of this form. Additionally, $L(z) = z(a+bi)$ for all $z \in \mathbb{C}$. Apparently this is the linalg version of the Cauchy-Riemann equation.

Analysis review!!! I havent taken analysis but i bet 10 bucks i know what hes about to review. Let $U \subseteq \mathbb{R}^n$ be open. Recall $f : U \to \mathbb{R}^m$ is *differentiable* at a point $x \in U$ if and only if there exists a linear transformation $L : \mathbb{R}^n \to \mathbb{R}^m$ and a map $\varphi : \mathbb{R}^n \to \mathbb{R}^m$ such that $f(x+h) = f(x) + L(h) + \|h\| \varphi(h)$ for $\|h\|$ in some ε-neighborhood of 0, and

$$\lim_{h \to 0} \varphi(h) = 0.$$

. Intuitively f is well-approximated by a linear function sufficiently close to a point x. I guess owe someone 10 bucks now. If this holds, L is called the *derivative* of f at x, also called the *Jacobian* $f_*(x) = \partial_i f^j$. It is simple to show that L is unique. Recall that matrix indicies are row then column.

The main subject of this course is *holomorphy* and holomorphic functions. In $\mathbb{R}^n$, it is interesting to say whether a function is differentiable at a point. In $\mathbb{C}$, this is not interesting at all and it is more interesting to consider holomorphy on open subsets.

Definition 1.1.1. Let $U \subseteq \mathbb{C}$ open. A function $f : U \to \mathbb{C}$ is called *holomorphic* on all of U if for all $z \in U$, when viewed as a map $f : U \subseteq \mathbb{R}^2 \to \mathbb{R}^2$ is differentiable at z and the Jacobian is $\mathbb{C}$-linear, meaning

$$\begin{pmatrix} \partial_x f^x & \partial_x f^y \\ \partial_y f^x & \partial_y f^y \end{pmatrix} \text{ is of the form } \begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

In particular, $\partial_x f^x = \partial_y f^y$ and $\partial_x f^y = -\partial_y f^x$. These are the *Cauchy-Riemann* equations, usually written as $f^x = u$ and $f^y = v$.

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}}$$

Thus definition 1.1.1 can be rewritten as a function which is everywhere differentiable on U and satisfies the Cauchy-Riemann equations.

Example 1.1.2. Let $f(z) = z$, so $u = x$ and $v = y$. Then $\partial_i f^j = \delta_i^j$, which satisfies the Cauchy-Riemann equations.

Let $f(z) = z^2$. Then

$$f(x + iy) = (x^2 - y^2) + i2xy.$$

The derivatives are

$$\frac{\partial u}{\partial x} = 2x, \qquad \frac{\partial v}{\partial y} = 2x, \qquad \frac{\partial v}{\partial x} = 2y, \qquad \frac{\partial u}{\partial y} = -2y.$$

This satisfies the Cauchy-Riemann equations.

Let $e^z = e^x (\cos y + i \sin y)$. Then

$$\partial_i f^j = \begin{pmatrix} e^x \cos y & e^x \sin y \\ -e^x \sin y & e^x \cos y \end{pmatrix}.$$

Let $f(z) = \overline{z}$. Then $\partial_i f^j = \sigma_z$ (easy to check) which does *not* satisfy Cauchy-Riemann.

Lecture 2: The complex differential operators

Thu 22 Jan 2026 09:32

By convention, we often identify the Jacobian matrix f_* of a holomorphic function f with the single complex number $f_*(1) \cdot (1, i)$. As explored last time, we can precisely recover f_* from such a number.

We now want to work in a more natural basis of differential operators for our tangent space, in particular ∂_z and $\partial_{\overline{z}}$. We claim these are indeed linearly independent, and thus form a basis since they are of the same cardinality as ∂_x, ∂_y. We define them as follows.

Definition 1.1.3. Define the differential operators

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right) \qquad \frac{\partial}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right).$$

This makes sense for several reasons.

Proposition 1.1.4. *A function $f : U \subseteq \mathbb{C} \to \mathbb{C}$ satisfies the C-R equations if and only if $\partial_{\overline{z}} f = 0$.*

Proof. Let $f = u + iv$. Then

$$\begin{aligned} \frac{\partial f}{\partial \overline{z}} &= \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right) \\ &= \frac{1}{2}\left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} + i\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y}\right) \\ &= \frac{1}{2}\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) + \frac{i}{2}\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right). \end{aligned}$$

By linear independence of $\{\partial_x, \partial_y\}$, this is zero if and only if the C-R equations are satisfied. ■

Proposition 1.1.5. *Let $f : U \subseteq \mathbb{C} \to \mathbb{C}$ be holomorphic. Then $f'(z) = \partial_z f|_z$.*

Proof. This follows immediately by the calculation done in the proof of proposition 1.1.4. ■

We then see by example 1.1.2 and proposition 1.1.5 that $\partial_z \overline{z} = 0$ and $\partial_{\overline{z}} z = 0$. We also have $\partial_z z = \partial_{\overline{z}} \overline{z} = 1$.

We now define a non-standard notion introduced by the book.

Definition 1.1.6. Let some $f : U \subseteq \mathbb{C} \to \mathbb{C}$. We will call f C^1-holomorphic if f is C^1 as a map $U \subseteq \mathbb{R}^2 \to \mathbb{R}^2$, and $\partial_{\overline{z}} f = 0$.

The only difference between definition 1.1.1 and definition 1.1.6 is that a C^1-holomorphic function has *continuous* partials. The more classical notion is the weaker notion. However, we will eventually see that C^1-holomorphic functions are equivalent to holomorphic functions. We will more precisely see that holomorphic functions are C^ω.

There is an even more traditional way of saying something is holomorphic.

Definition 1.1.7. Let $f : U \subseteq \mathbb{C} \to \mathbb{C}$. We say f is *holomorphic* if for all $z \in U$, the limit

$$f'(z) := \lim_{h \to 0 \text{ in } \mathbb{C}} \frac{f(z+h) - f(z)}{h}$$

exists.

I went afk for a bit but we ended up with this being equivalent to

$$f'(z) = \lim_{\mathbb{R} \ni t \to 0} \frac{f(z+it) - f(z)}{it}.$$

Since we are able to formulate derivatives in terms of the same formula used in calculus, the same proofs are used to prove derivative rules for holomorphic functions, such as

$$(fg)' = f'g + fg', \qquad (f \circ g)' = (f' \circ g) \cdot g'.$$

Definition 1.1.8. We say $f : U \subseteq \mathbb{C} \to \mathbb{C}$ is *analytic* if it locally can be given by a power series, meaning for all $z \in U$ there exists an open neighborhood $z \in V \subseteq U$ such that f has a power series expansion around z which converges in V. We can write V as an ε-neighborhood $B_\varepsilon(z)$ for some $\varepsilon > 0$ sufficiently small such that $B_\varepsilon(z) \subseteq U$. We say f is C^ω and write $f \in C^\omega(U, \mathbb{C})$.

A major goal of the next few weeks is to show that f is holomorphic on U if and only if f is analytic on U. This will imply that f is C^∞, and thus reconciling the definition given by [GK25].

Lecture 3: Analytic implies holomorphic

Tue 27 Jan 2026 09:34

Definition 1.1.9. Let $f : U \subseteq \mathbb{C} \to \mathbb{C}$. We say f is *analytic* or C^ω if for each $z \in U$, there exists an open set $x \in V \subseteq U$ such taht f admits a power series expansion which converges in V.

Proposition 1.1.10. *Suppose $\sum_n a_n w^n$ converges for some $\{a_n\}_{n\in\mathbb{N}} \subseteq \mathbb{C}$, $w \in \mathbb{C}$. If $z \in \mathbb{C}$ and $|z| < |w|$, then $\sum_n |a_n| |z|^n$ converges.*

Proof. Suppose z exists, meaning $w \neq 0$. Then

$$|a_n| |z|^n = |a_n| |w|^n \left(\frac{|z|}{|w|}\right)^n .$$

It follows kinda easily from here by noting that $\{a_n w^n\}_{n\in\mathbb{N}}$ is bounded. ∎

Corollary 1.1.11. *If $\sum_n a_n(z - z_0)^n$ converges for some $z \in B_r(z_0)$, $r > 0$, then it converges absolutely in the entire ball.*

Definition 1.1.12. Given a power series $\sum_n a_n(z - z_0)^n$, the *radius of convergence* is the number $R \in [0, \infty) \cup \{\infty\}$ such that the series converges for $z \in B_R(z_0)$, and diverges for $z \in \mathbb{C} \setminus \overline{B_R(z_0)}$.

Equivalently, $R = \sup\{|z - z_0| \mid \sum_n a_n(z - z_0)^n \text{ converges}\}$. Equivalently, recall that the lim sup of a sequence $\{c_n\}_{n\in\mathbb{N}} \subseteq \mathbb{R}$, $c_n \geq 0$ is

$$\limsup_{n\to\infty} c_n = \lim_{n\to\infty} \sup\{c_m\}_{m\geq n} .$$

We can then define $R = \left(\limsup_{n\to\infty} |a_n|^{1/n}\right)^{-1}$. It follows from these facts that if $a_n(z - z_0)^n$ converges, then so does replacing $a_n \to na_n$ or a_n/n.

Theorem 1.1.13. *Let f be analytic on $U \subseteq \mathbb{C}$. Then f is holomorphic on U.*

Proof. This is a nonstandard proof the professor came up with. Let $z_0 \in U$; it suffices to show $\partial_z f$ exists at z_0. Since f is analytic, there exists an open (wrt $\mathbb{C}$) neighborhood $V \subseteq U$ of z_0. Write

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n.$$

Let h sufficiently small such that the perturbation $f(z_0 + h)$ can be evaluated by this power series. Then

$$\frac{f(z_0 + h) - f(z_0)}{h} = \frac{\sum_{n\geq 0} a_n h^n - a_0}{h} = \frac{\sum_{n\geq 1} a_n h^n}{h} = \sum_{n\geq 1} a_n h^{n-1}.$$

The limit of the right hand side should be a_1. This makes sense since if we recall Taylor series, we should always find $a_n = f^{(n)}(z_0)/n!$. We need to show this limit exists, and we cannot assume power series are continuous on $\mathbb{C}$. Choose $|h| < r' <$ radius of convergence. Since it converges, there is ε such that

$$\sum_{n\geq m+1} |a_n| (r')^{n-1} < \frac{\varepsilon}{2}.$$

Since the partial sums are continuous functions in h, we see that they approach their limit, so it like follows. ∎

This proof was aids and I stopped following. There is a more canonical proof which also shows that
$$\partial_z f(z) = \sum_{n\geq 1} n a_n (z - z_0)^{n-1},$$
and that this also converges on an open disk that the original power series converges on.

Lecture 4: dsfghjk

Thu 29 Jan 2026 09:33

Proposition 1.1.14. *Let $\sum_n a_n(z - z_0)^n$ converge on some open disk $B_r(z_0)$. THen the function $f(z) = \sum_n a_n(z - z_0)^n$ is analytic on $B_r(z_0)$.*

Lecture 5: Harmonic functions

Tue 03 Feb 2026 09:31

Definition 1.1.15. A *harmonic function* is a C^2 function $u : U \subseteq \mathbb{R}^n \to \mathbb{R}$ such that the wave equation $\Delta u = \mathrm{Div}_\omega \, \mathrm{d}u = 0$ is satisfied.

Remark 1.1.16. Since vector fields admit a canonical multiplication operation, and since $[\partial_x, \partial_y] = 0$,
$$\frac{\partial}{\partial z}\frac{\partial}{\partial \overline{z}} = \frac{1}{4}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right) = \frac{1}{4}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) = \frac{1}{4}\Delta.$$

Proposition 1.1.17. *The real and imaginary parts of a holomorphic function $f = u + iv$ are harmonic.*

Proof. We have
$$\Delta f = 4\frac{\partial}{\partial z}\frac{\partial}{\partial \overline{z}} f = 4\frac{\partial}{\partial z} 0 = 0.$$
By linearity, $\Delta u + i\Delta v = 0$. By $\mathbb{R}$-linear independence of $\{1, i\}$ and since u, v are purely real, we have $\Delta u = \Delta v = 0$. ∎

This statement has a partial converse. Locally, we can take an open set to always be simply connected, hence giving us a local converse.

Proposition 1.1.18. *Let $u : U \subseteq \mathbb{R}^2 \to \mathbb{R}$ be harmonic, with U open and simply connected. Then there exists a holomorphic function $f : U \to \mathbb{C}$ such that $f = u + iv$ for some v.*

The proof requires a lemma from multivariate calculus.

Proposition 1.1.19. *Let $P, Q : U \subseteq \mathbb{R}^2 \to \mathbb{R}$ be C^1 on an open set U, and satisfy $\partial_x Q = \partial_y P$. Then there exists a C^2 function $\phi : U \to \mathbb{R}$ such that $\partial_x \phi$ and $\partial_y \phi$. Equivalently, $\mathrm{d}\phi = (P, Q)$, meaning ϕ is a* potential function *for the exact vector field $(P, Q) \in \mathfrak{X}(U)$.*

Proof of proposition 1.1.18. Define
$$P \coloneqq -\frac{\partial u}{\partial y}, \qquad Q \coloneqq \frac{\partial u}{\partial x}.$$
Then since u is harmonic, we have
$$-\frac{\partial P}{\partial y} + \frac{\partial Q}{\partial x} = -\left(-\frac{\partial^2 u}{\partial y^2}\right) + \frac{\partial^2 u}{\partial x^2} = \Delta u = 0.$$

Thus proposition 1.1.19 yields the potential function $v : U \to \mathbb{R}$ with $\mathrm{d}v = (-\partial_y u, \partial_x u)$. Note that this yields

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \qquad \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x}.$$

These are the C-R equations, so $f = u + iv$ is holomorphic on U. ∎

Chapter 2

Complex line integrals

Let $\gamma : [a, b] \to \mathbb{C}$. Then γ is C^1 if

- $\gamma|(a, b)$ is C^1, and
- $\partial_t \gamma$ extends to a continuous function on $[a, b]$.

Now let f be holomorphic on $U \subseteq \mathbb{C}$ and let $\operatorname{Im} \gamma \subseteq U$. Then we define

$$\int_\gamma f = \int_a^b f(\gamma(t))\gamma'(t)\,\mathrm{d}t.$$

We can view f as a section of the complexified cotangent bundle $T^*_{\mathbb{C}}U$. If $z = x + iy$, then we define $\mathrm{d}z = \mathrm{d}x + i\mathrm{d}y$. Then define the 1-form $\omega = f(z)\mathrm{d}z$. Recall that complex differentials make sense in the complexified (co)tangent bundle, so this is indeed a 1-form. We then have that our definition above yields

$$\int_\gamma f = \int_\gamma \omega = \int_a^b \gamma^*\omega.$$

As usual, this is the same formula:

$$\int_a^b \gamma^*\omega = \int_a^b \gamma^*(f(z)\mathrm{d}z).$$

Here we apply [Lee00] 11.25, which says if u is smooth and ω is a covector field, then the pullback along any F is $F^*(u\omega) = (u \circ F)F^*\omega$. This yields

$$= \int_a^b f(\gamma(t))\gamma^*\mathrm{d}z.$$

Note that $\gamma^*\mathrm{d}z$ is a 1-form on $[a, b]$. To see what it is, we see how it acts on the vector field ∂_t. We have $\gamma^*\mathrm{d}z(\partial_t) = \mathrm{d}z(\gamma_*\partial_t)$. The (real) pushforward of $\gamma(t) = \gamma^x(t) + i\gamma^y(t)$ is

$$\gamma_* = \begin{pmatrix} \partial_t \gamma^x \\ \partial_t \gamma^y \end{pmatrix} = \frac{\partial \gamma^x}{\partial t}\frac{\partial}{\partial x} + \frac{\partial \gamma^y}{\partial t}\frac{\partial}{\partial y}.$$

Acting this on ∂_t yields simply γ_*. Then acting $\mathrm{d}z$ on this yields

$$(\mathrm{d}x + i\mathrm{d}y)\left(\frac{\partial \gamma^x}{\partial t}\frac{\partial}{\partial x} + \frac{\partial \gamma^y}{\partial t}\frac{\partial}{\partial y}\right) = \frac{\partial \gamma^x}{\partial t} + i\frac{\partial \gamma^y}{\partial t} = \gamma'(t).$$

Since $\gamma^*\mathrm{d}z$ is a 1-form on $[a, b]$, it must be of the form $g(t)\mathrm{d}t$ for some g. By this computation, $\gamma^*\mathrm{d}z = \gamma'(t)\mathrm{d}t$. Note that this generalizes to piecewise-smooth curves γ.

Lecture 6: Complex line integrals

Tue 10 Feb 2026 09:32

Definition 2.0.1. Let $f : U \to \mathbb{C}$ be holomorphic. If $g : U \to \mathbb{C}$ is an *antiderivative* of f in the sense that $g' = f$, then we call g a *primitive* of f.

Terminology 2.0.2. Given $R \subseteq \mathbb{C}$ *closed*, we say f is holomorphic on R if it is holomorphic on some open neighborhood of R.

Theorem 2.0.3 (Goursat)**.** *Let $R \subseteq \mathbb{C}$ be the closed rectangle $[a, b] \times i[c, d]$, and let f be holomorphic on R. Then*

$$\int_{\partial R} f = 0.$$

Proof. Choose a piecewise smooth parameterization γ for ∂R. Later we will show that integrals are homotopy-invariant, so it makes sense that any parameterization will suffice. Write $\alpha = \int_{\partial R} f$, and assume for contradiction that $\alpha \neq 0$. The boundary ∂R inherits the Stokes orientation. Subdivide R into four isotopic rectangles R_i (bisect each side of R with a horizontal/vertical line). These are submanifolds which inherit orientations orientations via restricting the orientation form, and their boundaries inherit the Stokes orientation. These orientations conflict on the bisectors, but agree on the boundary. Hence they cancel on the interior, so

$$\left(\int_{\partial R_1} + \int_{\partial R_2} + \int_{\partial R_3} + \int_{\partial R_4}\right) f = \int_{\partial R} f.$$

We claim that there exists i with

$$\left|\int_{\partial R_i} f\right| \geq \frac{|\alpha|}{4}.$$

This follows immediately by

$$|\alpha| \leq \left|\sum_i \int_{\partial R_i} f\right| \leq \sum_i \left|\int_{\partial R_i} f\right| < |\alpha| .$$

Write $R^0 = R$ and R^1 for a choice of a subrectangle R_i with $\left|\int_{\partial R_i} f\right| \geq \frac{|\alpha|}{4}$. We can iterate this process to obtain a nested family of proper inclusions

$$R^0 \supsetneq R^1 \supsetneq \cdots .$$

Note that for all i,

$$\left|\int_{\partial R^i} f\right| \geq \frac{|\alpha|}{4^n}.$$

Let P_n be the length (perimeter) of ∂R^n. Let D_n be the diameter of R^n, computed by taking the line of an affine line through two corners of R^n. Note that $\lim \{R^n\} \cong *$, and thus $P_n, D_n \to 0$. So for any choice $z_n \in R^n$, the sequence $\{z_n\}$ is Cauchy. Since f is holomorphic, we can fix z_0 and choose sufficiently small h such that $f(z_0 + h) = f(z_0) + f'(z_0)h + |h| \varphi(h)$, for

$$\lim_{h\to 0} \varphi = 0.$$

Some nice intuition that I didn't write down earlier: φ is the higher order terms of the Taylor expansion of f, hence we are really considering a germ of a Taylor expansion. Take z sufficiently close to z_0 such that we may write $h = z - z_0$ with $z \to z_0$. Then this yields

$$f(z) = f(z_0) + f'(z-0)(z-z_0) + |z-z_0|\,\varphi(z-z_0).$$

Then

$$\int_{\partial R^n} f = f(z_0)\int_{\partial R^n} \mathrm{d}z + f'(z_0)\int_{\partial R^n}(z-z_0)\mathrm{d}z + \int_{\partial R^n} |z-z_0|\,\varphi(z-z_0)\mathrm{d}z.$$

Since 1 has a primitive and is the integral around a closed curve, we can apply the fundamental theorem of line integrals to see that $\int_{\partial R^n} \mathrm{d}z = 0$. For the same reason, the second integral vanishes. We then obtain

$$\int_{\partial R^n} f = \int_{\partial R^n} |z-z_0|\,\varphi(z-z_0)\mathrm{d}z \geq \frac{|\alpha|}{4^n}.$$

However,

$$\int_{\partial R^n} |z-z_0|\,\varphi(z-z_0)\mathrm{d}z \leq \int_{\partial R^n} |z-z_0|\,|\varphi(z-z_0)|\,\mathrm{d}z \leq D_n\varepsilon P_n = \frac{D}{\alpha^n}\frac{P}{\alpha^n}\varepsilon$$

for sufficiently small ε and sufficiently large n. We choose $\varepsilon = \frac{|\alpha|}{P_n D_n}$ and some cancelling yields the contradiction

$$\frac{|\alpha|}{4^n} < \frac{\alpha}{4^n}.$$

■

Remark 2.0.4. This did not require that f have a primitive. We will show next time that holomorphic functions always have primitives.

Lecture 7: Primitive along a path

Thu 26 Feb 2026 09:36

Let f be holomorphic on U. We currently only have integrals defined over piecewise C^1 paths, but to take advantage of the notion of homotopy we want to define them for C^0 paths. So let $\gamma : I \to U$ be C^0. We choose a partition $a = t_0 < t_1 < \cdots < t_n = b$ such that $\gamma[t_{i-1}, t_i]$ falls inside a disk D_i for all i. Since f is holomorphic, it has a primitive on any open disc, so let F_i be a primitive of f on D_i. Since $D_i \cup D_{i+1}$ is connected, the primitives F_i, F_{i+1} differ by at most a constant. We choose these constants such that $F_i = F_{i+1}$; since n is finite we obtain that $F_i = F_j$ for all i, j. For $t \in [t_i, t_{i+1}]$, define $F_\gamma(t) = F_i(\gamma(t)) = ((\bigcup_i F_i)\circ\gamma)(t)$. We define

$$\int_\gamma f \coloneqq F_\gamma(b) - F_\gamma(a).$$

Definition 2.0.5. A *primitive* F_γ along a path γ for a function $f \in \mathcal{O}(U)$ is a continuous function $F_\gamma : I \to \mathbb{C}$ with the property that for every $\tau \in I$, there is an open disk $D_\tau \subseteq U$ with $\tau \in D_\tau$ such that for t sufficiently close to τ, we have $F_\gamma(t) = F_\tau(\gamma(t))$ where F_τ is a primitive of f on D_τ.

Theorem 2.0.6. *Let $f \in \mathcal{O}(U)$ and γ, δ homotopic paths in U. Then*

$$\int_\gamma f = \int_\delta f.$$

Corollary 2.0.7 (Cauchy)**.** *Let $f \in \mathcal{O}(U)$ and γ be null-homotopic in U. Then*

$$\int_\gamma f = 0.$$

Lecture 8: Homotopy invariance

Tue 03 Mar 2026 09:31

Theorem 2.0.8. *Let $f : U \to \mathbb{C}$ be holomorphic, and let $\gamma, \delta : I \to U$ be continuous paths in U with the same endpoints, which are homotopic with the endpoints fixed. Then $\int_\gamma f = \int_\delta f$.*

Proof. Just note that if f is holo then $f\,\mathrm{d}z$ is closed, pull back along the homotopy, and throw Stokes at it. ∎

Corollary 2.0.9 (Cauchy integration formula)**.** *Let $f \in \mathcal{O}(U)$ and $z_0 \in U$ a point such that $\overline{B_r(z_0)} \subseteq U$. Then for any $z \in D \coloneqq D_r(z_0)$,*

$$f(z) = \frac{1}{2\pi i} \oint_{\partial D} \frac{f(\zeta)}{\zeta - z} \, \mathrm{d}\zeta.$$

Chapter 3

???

Lecture 9: Hai

Tue 24 Mar 2026 09:33

Definition 3.0.1. Let X be a space. We say X is simply connected if it is path connected and all loops are nullhomotopic, meaning $\pi_1(X) \cong 0$. Equivalently, $\Pi_1(X)(x,y)$ is always a singleton for any $x, y \in X$.

Theorem 3.0.2 (Riemann mapping)**.** *Every simply connected open subset $U \subseteq \mathbb{C}$ is biholomorphic to either the unit disk $D_1(0)$ or $\mathbb{C}$. However, there is no biholomorphism between $D_1(0)$ and $\mathbb{C}$.*

Remark 3.0.3 (Uniformization theorem)**.** Theorem 3.0.2 extends to Riemann surfaces: the only other simply connected Riemann surface up to isomorphism is the Riemann sphere $\mathbb{P}^1$.

Construction 3.0.4 (Riemann sphere)**.** Define the equivalence relation $\sim$ on $\mathbb{C}^2 \setminus 0$ by $z \sim \lambda z$ for all $z \in \mathbb{C}^2$ and $\lambda \in \mathbb{C}$. Hence we identify all complex 1-dimensional linear subspaces of $\mathbb{C}^2$, minus the origin. The quotient $\mathbb{P}^1 := (\mathbb{C}^2 \setminus 0) \sim$ is called the *Riemann sphere*. It is a complex manifold that is isomorphic to the complex 2-sphere S^2, and has charts given as follows. Given $(z_0, z_1) \in \mathbb{C}^2$, write $[z_0 : z_1]$ for the class of this point in $\mathbb{P}^1$. Then for $i = 0, 1$, define

$$U_i = \{[z_0 : z_1] \mid z_i \neq 0\}, \qquad \varphi : U_i \to V \subseteq \mathbb{C}, \qquad \varphi_0[z_0 : z_1] = z_1, \qquad \varphi_1[z_0 : z_1] = z_0.$$

The set U_i is well-defined since $[z_0 : z_1] = \lambda[z_0 : z_1]$, and if we pull out any scalar while z_i is 0 then it stays 0. The Riemann sphere is compact and is isomorphic to $\mathbb{C} \cup \{\infty\}$.

Recall that $f \in \mathcal{O}(U)$ can be viewed as a smooth map $f : U \subseteq \mathbb{R}^2 \to \mathbb{R}^2$, and its derivative has the form

$$\mathrm{D}f = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

Consider the map $f(z = a + ib) = 1/\sqrt{a^2 + b^2}$. Then

$$\mathrm{D}f = \begin{pmatrix} \frac{a}{\sqrt{a^2+b^2}} & \frac{-b}{\sqrt{a^2+b^2}} \\ \frac{b}{\sqrt{a^2+b^2}} & \frac{a}{\sqrt{a^2+b^2}} \end{pmatrix}.$$

This is both of the form of a holomorphic function, *and* is orthogonal.

A conformal map is one which preserves angles.

Lecture 10: Analytic continuation

Thu 26 Mar 2026 09:34

Let f be holomorphic on $V \subseteq \mathbb{C}$ and let $U \subseteq \mathbb{C}$ be a connected open set which properly contains V. The question is whether f extends to a holomorphic function on U. In general it is impossible to say; it depends on f, V, and U. The principle of analytic continuation tells us that if f *does* extend, then the extension is unique.

Proposition 3.0.5. *Let U be a connected open subset of $\mathbb{C}$, and let $f, g : U \to \mathbb{C}$ be holomorphic. If there is an infinite compact subset $K \subseteq U$ such that $f|K = g|K$, then $f = g$.*

In particular, let h be holomorphic on some open V, and choose a compact subset $K \subseteq V$. If f, g are holomorphic extensions of h to a connected open set U containing V, then $f|V = g|V = h$. In particular, $f|K = g|K$, implying that $f = g$. This statement is absolutely false for C^∞ functions: suppose we have a function $h : V \to \mathbb{C}$ compactly supported in V. Then one can extend it to f on U with $V \subseteq U$ by taking either $f|(U \setminus V) = 0$, or by taking f to be some bump function supported in U.

We can generalize somewhat.

Lemma 3.0.6. *Let D be an open ball of radius r about z_0, and suppose h is holomorphic on D. Suppose that $\{z_n\}_{n\geq 1}$ is a sequence in d with $z_n \to z_0$ such that $z_n \neq z_0$ for infinitely many n. Assume that $h(z_n) = 0$ for $n \geq 1$. Then $h = 0$.*

Proof. Since $h \in \mathcal{O}(D)$, we have $h \in C^0(D)$, and thus

$$h\left(\lim_{n\to\infty} z_n\right) = \lim_{n\to\infty} h(z_n) = \lim_{n\to\infty} 0 = 0.$$

We now have a countable set $Z := \{z_n\}_{n\geq 0} \subseteq D$ such that $h|Z = 0$.

Assume for contradiction that in a neighborhood of z_0, we can write

$$h(z) = \sum_{n\geq 0} a_n (z - z_0)^n, \qquad z \in D$$

for a_n not all zero. Let m be the smallest integer such that $a_m \neq 0$. We then write

$$h(z) = \sum_{n\geq m} a_n (z - z_0)^n = (z - z_0)^m \sum_{n\geq m} a_n (z - z_0)^{n-m}.$$

Write

$$\ell(z) = \sum_{n\geq m} a_n (z - z_0)^{n-m}.$$

Then ℓ is holomorphic on some neighborhood of z_0. Note also that $\ell(z_0) = a_m \neq 0$. Continuity implies that there is $\varepsilon > 0$ sufficiently small that ℓ is nonzero in an ε-ball $D_\varepsilon(z_0)$ of z_0. By definition of convergence, there is N sufficiently large such that $n \geq N$ implies $z_n \in D_\varepsilon(z_0)$. Then

$$h(z_0) = (z_n - z_0)^m \ell(z_n).$$

We have $z_n \neq z_0$ for infinitely many n, so we can choose $n \geq N$ such that $z_n \neq z_0$. Thus the first term is nonzero. By the same logic, $\ell(z_n)$ is nonzero since at least one coefficient a_m is nonzero. Thus $h(z_n)$ is nonzero, a contradiction. ■

Proof of analytic continuation. Let U be connected, $K \subseteq U$ an infinite compact set, and $f, g \in \mathcal{O}(U)$ such that $f|K = g|K$. Let $h = f - g$ such that $h|K = 0$; it suffices to show $h = 0$.

Let $A \subseteq U$ be the set of all $z_0 \in U$ such that h vanishes in some neighborhood of z_0; by definition A is open. Let $B = U \setminus A$. We show that B is open and A is nonempty. With these statements shown, if B were nonempty, we then realize U as a union $A \cup B$ of disjoint nonempty open sets, contradicting connectedness. It suffices to show A is closed.

We first show A is closed. It suffices to show that if $\{z_n\}$ is a sequence with $z_n \to z_0$, $z_0 \in U$, then $z_0 \in A$. Sketch: choose some $D_\varepsilon(z_0) \subseteq U$, choose z_n in K, apply lemma, and get $h|D_\varepsilon(z_0) = 0$.

Finally, we use the fact that compactness and sequential compactness coencide in metric spaces. Choose a sequence in K; we have $h|\{x_n\}_{n\geq 0}$. There is a convergent subsequence, which converges to some z_0. By the lemma, $h(z_0) = 0$. ∎

Lecture 11: The principle of analytic continuation

Tue 31 Mar 2026 09:34

Example 3.0.7. We cna show that $\exp z$ and

$$\log z = \sum_{n\geq 1} (-1)^{n+1} \frac{(z-1)^n}{n}, \qquad |z-1| < 1$$

satisfy the relation $\exp(\log z) = z$ by sorting out the power series. We can also show the fllowing: let $f(z) = e^{\ln z}$. Sufficiently close to $z = 1$, we can show that $g(z) = e^{\ln z} = z$. Thus we have $f|B = g|B$ sufficiently close to $z = 1$, so the principle of analytic continuation yields $e^{\ln z} = 1$ on a disk $D_1(1)$. We cannot go past this due to ln having a pole at 0.

Theorem 3.0.8 (Morena)**.** *Let f be* continuous *on U and suppose that for all rectangles $R \subseteq U$,*

$$\int_{\partial R} f = 0.$$

Then f is holomorphic.

Proof. Suffices to show that f is holomorphic on any open disk in U. Let $D = D_r(z_0) \subseteq u$. We show f is holomorphic on D.

Define $F : D \to \mathbb{C}$ by

$$F(z) = \int_{\gamma_z} f,$$

where γ_z is a path $z_0 \to z$ for some $z \in D$. Since D is contractible and simply connected, all such paths are homotopic, and at least one such path exists. In particular, let R be a rectangle with z_0 and z the opposite corners. Then take two paths γ_z, δ_z to both travel one side of the boundary of the rectangle. Reversing the orientation of one of them and concatenating yields a parameterization of ∂R. We then have

$$0 = \int_{\partial R} f = \int_{-\delta_z} f + \int_{\gamma_z} f.$$

One may follow the same method as in the proof to Goursatz to find a local primitive f with $F'(z) = f(z)$, showing F is holomorphic. By the Cauchy Integral formula, the derivative of F is holomorphic. Therefore f is holomorphic since it is locally a primitive. ∎

Theorem 3.0.9. *Let $\{f_n\}_{n\geq 1}$ be a sequence of functions on U which is uniformly convergent on compact sets in U to a function f on U. I.e. for all compact $K \subseteq U$, the sequence converges on restrictions $f_n|K \to f|K$. Then f is holomorphic on U.*

Proof. We know that $f|K$ is continuous for all compact subsets K, we can take $K = \overline{D}$ for any disk $D \subseteq U$ to obtain that f is continuous on U.

Now fix D such that $\overline{D} \subseteq U$ and apply the hypothesis on $K = \overline{D}$ or $K = \partial R$ for a rectangle R. Then holomorphy yields

$$\int_{\partial R} f_n = 0 \qquad \forall n.$$

Uniform convergence yields

$$0 = \int_{\partial R} f_n|R \to \int_{\partial R} f|R = 0.$$

Therefore f is holomorphic by the above lemma. ■

Definition 3.0.10. Given a series $\sum_{n\geq 1} f_n$ of holomorphic functions, we say $\sum_{n\geq 1} f_n$ is *normally convergen on compact sets* in U if for each compact $K \subseteq U$, there is a sequence $\{M_n\}_{n\geq 1}$ of real numbers $M_n \geq 0$ such that

1. $|f_n(z)| \leq M_n$ on K, and
2. $\sum_{n\geq 1} M_n$ converges.

These are the hypotheses of the Weierstrass M-test, which tells us:

Theorem 3.0.11. *If $\sum_{n\geq 1} f_n$ converges normally on compact sets, then it converges uniformally on compact sets.*

Let $n > 0$ a real number. Write $n^s := e^{s \ln n}$ for $s \in \mathbb{C}$. For $\Re(s) > 1$, we define

$$\zeta(s) = \sum_{n\geq 1} n^{-s},$$

the Riemann ζ function. To see that ζ is holomorphic on $\Re(s) > 1$, we show it is normally convergent. It suffices to show it is normally convergent on $U_{1+\varepsilon}$ given by $\Re(s) > 1 + \varepsilon$ for $\varepsilon > 0$.

Let $M_n = n^{-(1+\varepsilon)}$. Then if $s \in K \subseteq U_{1+\varepsilon}$, get $\Re(s) \geq 1 + \varepsilon$, so

$$\left|n^{-s}\right| \leq n^{-(1+\varepsilon)}.$$

Thus we have our bounds, and the series trivially converges. Therefore ζ is holomorphic with $\Re(s) > 1$.

Fact: there is an analytic continuation $\zeta(s)$ to $\mathbb{C} \setminus \{1\}$, which means a function $f \in \mathcal{O}(\mathbb{C} \setminus \{1\})$ which agrees with ζ on the restriction $f|\{\Re(s) > 1\} = \zeta$. We define the Riemann zeta function as this continuation.

For now, we can demonstrate the analytic continuation to $\Re(s) > 0$, $s \neq 1$.

Lecture 12: a

Thu 02 Apr 2026 09:32

Series

Lecture 13: Winding number

Tue 14 Apr 2026 09:31

Let $f : U \subseteq \mathbb{R}^2 \to \mathbb{R}^2$ be C^1. Then recall that f has a local C^1 inverse if and only if $\mathrm{D}f$ is invertible. Now if f is holomorphic, then $\det \mathrm{D}f = a^2 + b^2$. Here $a = \partial_x u = \partial_y v$ and $b = \partial_y u = -\partial_x v$. This obviously happens if and only if $a + ib \neq 0$. Thus all nonzero holomorphic functions have local inverses. Moreover, this local inverse is holomorphic.

Definition 3.0.12. Given a piecewise C^1 closed curve $\gamma : I \to \mathbb{C}$, and given $z \in \mathbb{C} \setminus \gamma(I)$, define the *winding number* by

$$W_\gamma(z) = \frac{1}{2\pi i} \int_\gamma \frac{1}{\zeta - z} \mathrm{d}\zeta.$$

Remark 3.0.13. The curve γ defines a homotopy class in $\pi_1(\mathbb{C} \setminus \{z\})$. The map $[\gamma] \mapsto W_\gamma(z)$ is an isomorphism $\pi_1(\mathbb{C} \setminus \{z\})$. Of course, we know that such an isomorphism exists since $\mathbb{C} \setminus \{z\}$ has the homotopy type of S^1.

Of course remark 3.0.13 relies on

Proposition 3.0.14. *With notation as in definition 3.0.12, $W_\gamma(z) \in \mathbb{Z}$.*

Proof. The book's proof is quite computational, so we give a more conceptual proof. Recall that e^z is a local biholomorphism. Recall that

$$\frac{\mathrm{d}}{\mathrm{d}\zeta} \ln \zeta = \frac{1}{\zeta}.$$

Thus

$$\frac{\mathrm{d}}{\mathrm{d}\zeta} \ln(\zeta - z) = \frac{1}{\zeta - z}.$$

Choose a primitive F of $\frac{1}{\zeta - z}$ along γ. Thus

$$\int_\gamma \frac{\mathrm{d}\zeta}{\zeta - z} = F(\gamma(1)) - F(\gamma(0)).$$

Recall that a primitive is defined as a collection $\{F_j\}_{j \geq 0}$ of maps defined on some open subset $F_j : U_j \to \mathbb{C}$ with $\gamma(I) \cap U_j \neq \varnothing$, and such that the U_j's cover $\gamma(I)$. We generally order them in the obvious way.

Note that we can take $F_1(\zeta) = \ln(\zeta - z)$, such that $E^{F_1(\zeta)} = \zeta - z$. This relation must thus hold in the overlap $U_1 \cap U_2$, and the principle of analytic continuation implies $e^{F_2(\zeta)} = \zeta - z$. Induction proves this for all n. Thus

$$F(\gamma(1)) - F(\gamma(0)) = \ln(\zeta - z) - \ln(\zeta - z).$$

However, these logs can be defined on different branches, so this need not be zero. However, they are equal when taking the exponential, meaning they differ by an element of the kernel $\operatorname{Ker} \exp \cong \mathbb{Z}$. Thus this must be an element of $\mathbb{Z}$. ■

Theorem 3.0.15 (Cauchy Integral Formula, General Case)**.** *Let $f : U \to \mathbb{C}$ holomorphic and $\gamma : I \to U$ is a piecewise C^1 closed, null-homotopic curve. Let $z \in U \setminus \gamma(I)$. Then*

$$W_\gamma(z) f(z) = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{z - \zeta} \mathrm{d}\zeta.$$

Theorem 3.0.16 (Residue)**.** *Let $f : U \to \mathbb{C}$ be meromorphic with finitely many poles $\{z_i\}_{i\in[n]} \subseteq U$. Let $\gamma : U \to \mathbb{C}$ be a piecewise C^1 closed curve, which is null-homotopic on U. Assume that $\{z_i\}_{i\in[n]} \cap \gamma(I) = \varnothing$. Then*

$$\int_\gamma f = 2\pi i \sum_{0\le i\le n} W_\gamma(z_i) \operatorname{Res}_{z_i} f.$$

Example 3.0.17. Show

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2+1}\,\mathrm{d}x = \frac{\pi}{e}.$$

Let $f(z) = \frac{e^{iz}}{z^2+1}$ and consider the contouor γ_R which traverses the line segment $[-R, R] \subseteq \mathbb{R}$, and then traverses the semicircle of radius $R > 1$ counterclockwise from R to $-R$. The residues of f are $\pm i$, so for $R > 1$, there is a residue i inside the contour. By the Residue theorem,

$$\int_{\gamma_R} f = 2\pi i \cdot \operatorname{res}_i f.$$

Now we compute the residue. We factor

$$f(z) = \frac{\exp(iz)}{(z+i)(z-i)}.$$

The residue is thus

$$\operatorname{res}_i f = \lim_{z\to i}(z-i)f(z) = \lim_{z\to i}\frac{\exp(iz)}{z+i} = \frac{\exp(-1)}{2i} = \frac{1}{2ei}.$$

Thus

$$\int_{\gamma_R} f = \frac{2\pi i}{2ei} = \frac{\pi}{e}.$$

The integral is independent of R, and in particular we can take $R \to \infty$. Write C_R for the semicircle component of γ. We conclude by showing that $\int_{C_R} f \to 0$ in the limit as $R \to \infty$ (because $\sin x$ is odd so that part vanishes apparently). We compute

$$\lim_{R\to\infty}\int_{C_R} \frac{e^{iz}}{z^2+1}\mathrm{d}z.$$

This limit exists because it's sufficiently bounded for all $R > 1$. Well, if z is in the upper half plane $\operatorname{Im} z \ge 0$, then $\left|e^{iz}\right| < 1$. Thus the top part of the integral will vanish in the limit to the boundary of the upper half plane, so we estimate the denominator:

$$\left|z^2+1\right| \ge \left|z^2\right| - 1 = R^2 - 1 \ge \frac{R^2}{2}, \qquad R \ge \sqrt{2}.$$

Note that $\left|\frac{R^2}{2} \ge 1\right|$. Thus the denominator does not vanish but the numerator does, meaning that the integrand can be taken to 0 in the limit $R \to \infty$. One can rigorize this more but I do not care.

Example 3.0.18. Calculate

$$\int_0^{2\pi} \frac{\mathrm{d}\theta}{2+\sin\theta}.$$

Let U be connected and $M(U)$ the field of meromorphic functions on U. The *logarithmic derivative* $L : M(U)^\times \to M(U)$ is the map $L(f) = f'/f$. This is a group homomorphism:

$$L(fg) = \frac{f'g+g'f}{fg} = \frac{f'}{f} + \frac{g'}{g} = L(f) + L(g).$$

Lecture 14: Rouche

Thu 23 Apr 2026 09:34

Theorem 3.0.19 (Rouche)**.** *Let f, g be holomorphic on $\overline{D}$, where $D = B_r(z_0)$. Suppose that $|f - g| < |f|$ on ∂D. Then f and g have the same number of zeros, counted with mutliplicity, in D.*

The proof proceeds by integrating along the boundary and using the argument principle.

Theorem 3.0.20 (Open Mapping)**.** *Let f be a nonconstant holomorphic function on a connected open set U. Then f is open.*

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